Structures and Materials

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Home Task

1. Composites UD ply

Task 1: Given are the material properties of a UD ply made of C-AS4 EP.

$$E_1 = 142\ GPa \qquad E_2 = 10.3\ GPa \qquad G_{12} = 7200\ MPa \qquad \nu_{12} = 0.27$$

A) Calculate the compliance matrix of the UD ply.

$$\frac{\nu_{12}}{E_1} = \frac{\nu_{21}}{E_2} = \frac{0.27}{142000}$$

$$[S] = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{21}}{E_2} & 0 \\ -\frac{\nu_{12}}{E_1} & \frac{1}{E_2} & 0 \\ 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} = \begin{bmatrix} \frac{1}{142000} & -\frac{0.27}{142000} & 0 \\ -\frac{0.27}{142000} & \frac{1}{10300} & 0 \\ 0 & 0 & \frac{1}{7200} \end{bmatrix}$$

B.1) Calculate the stiffness matrix [Q] of the UD ply.

$$\nu_{21} = \frac{\nu_{12} E_2}{E_1} = \frac{0.27 \cdot 103}{142} = 0.0196$$

$$[Q] = \begin{bmatrix} \frac{E_1}{1-\nu_{12}\nu_{21}} & \frac{\nu_{12}E_2}{1-\nu_{12}\nu_{21}} & 0 \\ \frac{\nu_{21}E_1}{1-\nu_{12}\nu_{21}} & \frac{E_2}{1-\nu_{12}\nu_{21}} & 0 \\ 0 & 0 & G_{12} \end{bmatrix} = \begin{bmatrix} 142755.5 & 2795.8 & 0 \\ 2795.8 & 10354.8 & 0 \\ 0 & 0 & 7200 \end{bmatrix}$$

B.2) Calculate the stiffness matrix of the UD ply by matrix inversion of S.

$$[Q] = [S]^{-1} = \begin{bmatrix} 142754.9 & 2795.8 & 0 \\ 2795.8 & 10354.8 & 0 \\ 0 & 0 & 7200 \end{bmatrix}$$

using the calculator CASIO fx-991 ES numerical matrix inversion.

B.3) Calculate the stiffness matrix [Q] of the UD ply using elamX2.

2. Composites Laminate

Task 2: The UD ply from Task 1 is used to setup a laminate with the stacking sequence $[0 / 45]_S$. The ply thickness is $t = 0.125mm$

A.1) Calculate the stiffness matrix $[Q]_{45}$.

$$[\bar{Q}] = [T][Q][T]^T$$

$$[\boldsymbol{T}] = \begin{bmatrix} cos^2(45) & sin^2(45) & -2\sin(45)\cos(45) \\ sin^2(45) & cos^2(45) & 2\sin(45)\cos(45) \\ \sin(45)\cos(45) & -\sin(45)\cos(45) & cos^2(45) - sin^2(45) \end{bmatrix}$$

$$\cos(45) = \frac{\sqrt{2}}{2} \qquad \sin(45) = \frac{\sqrt{2}}{2}$$

$$[\boldsymbol{T}] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -1 \\ \frac{1}{2} & \frac{1}{2} & 1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix} \qquad [\boldsymbol{T}]^T = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -1 & 1 & 0 \end{bmatrix}$$

$$[\bar{Q}] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -1 \\ \frac{1}{2} & \frac{1}{2} & 1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 142755.5 & 2795.8 & 0 \\ 2795.8 & 10354.8 & 0 \\ 0 & 0 & 7200 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -1 & 1 & 0 \end{bmatrix}$$

$$[\bar{Q}] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -1 \\ \frac{1}{2} & \frac{1}{2} & 1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 142755.5 \cdot \frac{1}{2} + 2795.8 \cdot \frac{1}{2} & 142755.5 \cdot \frac{1}{2} + 2795.8 \cdot \frac{1}{2} & 0 \\ 2795.8 \cdot \frac{1}{2} + 10354.8 \cdot \frac{1}{2} & 2795.8 \cdot \frac{1}{2} + 10354.8 \cdot \frac{1}{2} & 0 \\ -7200 & 7200 & 0 \end{bmatrix}$$

$$[\bar{Q}]_{45} = \begin{bmatrix} 46875.5 & 32475.5 & 33100.2 \\ 32475.5 & 46875.5 & 33100.2 \\ 33100.2 & 33100.2 & 36879.7 \end{bmatrix}$$

A.2) Use your calculator to do the matrix multiplication.

A.3) Use elamX2 to derive the rotated matrix.